

7.28 Assume $\vec{E}(x, y, z, t) = (E_0(x, y) \vec{e}_1 \pm i E_0(x, y) \vec{e}_2 + E_3 \vec{e}_3) e^{ikz - i\omega t}$

Using $\nabla \cdot \vec{E} = 0$, we have $\frac{\partial E_0}{\partial x} \pm i \frac{\partial E_0}{\partial y} + ik E_3 = 0$, which leads to

$$E_3 = \frac{i}{k} \left(\frac{\partial E_0}{\partial x} \pm i \frac{\partial E_0}{\partial y} \right)$$

Given $\vec{E}$, from $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = i\omega \vec{B}$, assuming frequency ω , we have

$\vec{B} = -\frac{i}{\omega} \nabla \times \vec{E}$. In component form,

$$\begin{aligned} \nabla \times \vec{E} &= \begin{vmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_0 e^{ikz} & \pm i E_0 e^{ikz} & \frac{i}{k} \left(\frac{\partial E_0}{\partial x} \pm i \frac{\partial E_0}{\partial y} \right) e^{ikz} \end{vmatrix} e^{-i\omega t} \\ &= \left[\vec{e}_1 \left(\frac{i}{k} \left(\frac{\partial^2 E_0}{\partial x \partial y} \pm i \frac{\partial^2 E_0}{\partial y^2} \right) e^{ikz} + i E_0 \frac{\partial}{\partial z} e^{ikz} \right) + \vec{e}_2 \left(E_0 \frac{\partial}{\partial z} e^{ikz} - \frac{i}{k} \left(\frac{\partial^2 E_0}{\partial x^2} \pm \frac{\partial^2 E_0}{\partial x \partial y} \right) e^{ikz} \right) \right. \\ &\quad \left. + \vec{e}_3 \left(\pm i \frac{\partial E_0}{\partial x} - \frac{\partial E_0}{\partial y} \right) e^{ikz} \right] e^{-i\omega t} \end{aligned}$$

Ignoring second order derivatives, we will arrive at

$$\nabla \times \vec{E} = \left[\pm k E_0 \vec{e}_1 + ik E_0 \vec{e}_2 \pm i \left(\frac{\partial E_0}{\partial x} \pm i \frac{\partial E_0}{\partial y} \right) \vec{e}_3 \right] e^{ikz - i\omega t}$$

Then $\vec{B} = -\frac{i}{\omega} \nabla \times \vec{E} = \left[\mp \frac{ik}{\omega} E_0 \vec{e}_1 + \frac{k}{\omega} E_0 \vec{e}_2 \mp \frac{i}{\omega} \left(\frac{\partial E_0}{\partial x} \pm i \frac{\partial E_0}{\partial y} \right) \vec{e}_3 \right] e^{ikz - i\omega t}$

$$= \mp \frac{ik}{\omega} \left[E_0 \vec{e}_1 \pm i E_0 \vec{e}_2 + \frac{i}{k} \left(\frac{\partial E_0}{\partial x} \pm i \frac{\partial E_0}{\partial y} \right) \vec{e}_3 \right] e^{ikz - i\omega t}$$

$$= \mp i \sqrt{\mu \epsilon} \vec{E}$$

where in the last step we have use $\frac{k}{\omega} = \frac{n}{c} = \sqrt{\mu \epsilon}$